CPPDescent

229ddeb (with uncommitted changes)

Generated by Doxygen 1.9.6

Chapter 1

CPP-Descent

Part of the "Software Development for Computing Systems" course of the Department of Informatics and Telecommunications, UoA for the first semester of the academic year 2023-2024.

This project constitutes an implementation of the KNN-Graph creation and search algorithms described in [1], [2].

1.1 Code Structure

In the wiki, you can find documentation generated by Doxygen.

1.2 Contributors

- Konstantinos Chousos (1115202000215)
- Anastasios-Fedon Seitanidis (1115202000179)

1.3 Bibliography

[1] W. Dong, C. Moses, and K. Li, "Efficient k-nearest neighbor graph construction for generic similarity measures," in *Proceedings of the 20th international conference on World wide web*, in WWW '11. New York, NY, USA: Association for Computing Machinery, Mar. 2011, pp. 577–586. doi: 10.1145/1963405.1963487.

[2] "How PyNNDescent works — pynndescent 0.5.0 documentation." Accessed: Oct. 10, 2023. [Online]. Available: https://pynndescent.readthedocs.io/en/latest/how_pynndescent_works.html

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Chapter 3

File Index

3.1 File List

Here is a list of all documented files with brief descriptions:

Chapter 4

Class Documentation

4.1 cppdescent Class Reference

A sample function for the cppdescent class, used to test the correctness of the CMake files.

```
#include <cppdescent.hpp>
```

Public Member Functions

- int hello (void)

  Prints "Hello World".

4.1.1 Detailed Description

A sample function for the cppdescent class, used to test the correctness of the CMake files.

4.1.2 Member Function Documentation

4.1.2.1 hello()

```
int cppdescent::hello (
            void   )
```

Prints "Hello World".

Returns

int exit code

The documentation for this class was generated from the following files:

- cppdescent.hpp
- cppdescent.cpp

4.2 List Class Reference

Public Member Functions

- **List** (DestroyFunc destroyValue)
- int **getSize** ()
- void **insertNext** (ListNode *node, Pointer value)
- void **removeNext** (ListNode *node)
- ListNode **find** ()
- int **setDestroyValue** (int value)
- ListNode **next** (ListNode node)
- Pointer **nodeValue** (ListNode node)
- ListNode **findNode** (ListNode node)
- int **increaseSize** ()
- int **decreaseSize** ()
- ListNode * **getHead** ()
- ListNode * **getTail** ()

The documentation for this class was generated from the following files:

- linkedList.hpp
- linkedList.cpp

4.3 ListNode Class Reference

Public Member Functions

- **ListNode** (Pointer value)
- int **setNext** (ListNode *next)
- ListNode * **getNext** ()
- Pointer **getValue** ()

The documentation for this class was generated from the following files:

- linkedList.hpp
- listNode.cpp

Chapter 5

File Documentation

5.1 common.hpp

```
#pragma once

typedef void* Pointer;

typedef int (*CompareFunc)(Pointer a, Pointer b);

typedef void (*DestroyFunc)(Pointer value);
```

5.2 cppdescent.hpp

```

class cppdescent {
 public:
  int hello(void);
};
```

5.3 linkedList.hpp

```
#pragma once
#include "common.hpp"

class ListNode {
 public:
  ListNode();
  ListNode(Pointer value) : value(value){};
  ~ListNode();
  int setNext(ListNode* next);
  ListNode* getNext();
  Pointer getValue();

 private:
  ListNode* next;
  Pointer value;
};

class List {
 public:
  List(DestroyFunc destroyValue);
  ~List();
  int getSize();
  void insertNext(ListNode* node, Pointer value);
  void removeNext(ListNode* node);
  ListNode find();
  int setDestroyValue(int value);
  ListNode next(ListNode node);
  Pointer nodeValue(ListNode node);
  ListNode findNode(ListNode node);
  int increaseSize();
  int decreaseSize();
```

```
  ListNode* getHead();
  ListNode* getTail();

 private:
  ListNode* dummy;
  ListNode* head;
  ListNode* tail;
  int size;
  DestroyFunc destroyValue;
};
```

5.4 cppdescent.cpp File Reference

Implementation of cppdescent library.

```
#include "cppdescent/cppdescent.hpp"
#include <iostream>
```

Include dependency graph for cppdescent.cpp:

Index